JavaScript Comparison

Previous Next

Comparison Operators

Comparison operators are used to **compare two values**.

Comparison operators always return `true` or `false`.

Given that `x = 5`, the table below explains the comparison operators:

Operator	Description	Comparing	Returns	
==	equal to	x == 8	false	Try it »
		x == 5	true	Try it »
		x == "5"	true	Try it »
===	equal value and equal type	x === 5	true	Try it »
		x === "5"	false	Try it »
!=	not equal	x != 8	true	Try it »
!==	not equal value or not equal type	x !== 5	false	Try it »
		x !== "5"	true	Try it »
		x !== 8	true	Try it »
>	greater than	x > 8	false	Try it »
<	less than	x < 8	true	Try it »

Comparison operators can be used in conditional statements to compare values and take action depending on the result:

```
if (age < 18) text = "Too young to buy alcohol";
```

You will learn more about the use of conditional statements in the if...else chapter of this tutorial.

JavaScript String Comparison

All the comparison operators above can also be used on strings:

Example

```
let text1 = "A";
let text2 = "B";
let result = text1 < text2;
```

Try it Yourself »

Note that strings are compared alphabetically:

Example

```
let text1 = "20";
let text2 = "5";
let result = text1 < text2;
```

Try it Yourself »

Comparing data of different types may give unexpected results.

When comparing a string with a number, JavaScript will convert the string to a number when doing the comparison. An empty string converts to 0. A non-numeric string converts to `NaN` which is always `false`.

Case	Value	Try
2 < 12	true	Try it »
2 < "12"	true	Try it »
2 < "John"	false	Try it »
2 > "John"	false	Try it »
2 == "John"	false	Try it »
"2" < "12"	false	Try it »
"2" > "12"	true	Try it »
"2" == "12"	false	Try it »

When comparing two strings, "2" will be greater than "12".

Alphabetically 1 is less than 21.

To secure a proper result, variables should be converted to the proper type before comparison:

Example

```
age = Number(age);
if (isNaN(age)) {
  voteable = "Input is not a number";
} else {
  voteable = (age < 18) ? "Too young" : "Old enough";
}
```

Try it Yourself »